The exact discrete gradient of the average-vector-field method

Proof of Conjecture 00000007367

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Abstract

The interval average of the differential of a continuously differentiable energy is an exact discrete differential. It satisfies both the energy difference identity and consistency on the diagonal. Pairing this average with a skew Hamiltonian operator proves exact energy preservation of every average-vector-field step. We prove the full identity for arbitrary real normed spaces using genuine Fréchet derivatives and Bochner integrals, and verify it in Lean.

Precise statement and conventions

The bilingual source asserts the exact discrete-gradient identity and its realization by the averaging operator in the average-vector-field (AVF) method. We use its standard differentiable setting: an energy H with continuous first derivative, and exact integration in the definition of the method. No restriction to quadratic energies is needed.

Let E be a real normed vector space and let E^* denote its continuous dual. Suppose $H : E \to \mathbb{R}$ is differentiable and $DH : E \to E^*$ is continuous. For $x, y \in E$, define

$$\gamma_{x,y}(t) = x + t(y - x), \qquad \overline{D}H(x, y) = \int_0^1 DH(\gamma_{x,y}(t))\, dt \in E^*.$$

The integral is the Bochner integral. The dual E^* is complete even when E is not, and the integrand is continuous on the compact interval, so this integral exists.

In a finite-dimensional Euclidean space the Riesz identification gives the familiar averaged gradient

$$\overline{\nabla}H(x, y) = \int_0^1 \nabla H(x + t(y - x))\, dt, \qquad \overline{D}H(x, y)[v] = \langle \overline{\nabla}H(x, y), v\rangle.$$

Thus the covector formulation below is the ordinary AVF discrete gradient expressed without choosing coordinates or an inner product.

Theorem 1 (Exact discrete differential)**.** *For all $x, y \in E$,*

$$\overline{D}H(x, y)[y - x] = H(y) - H(x), \tag{1}$$

$$\overline{D}H(x, x) = DH(x). \tag{2}$$

Proof. The chain rule gives, for every real t,

$$\frac{d}{dt} H(\gamma_{x,y}(t)) = DH(\gamma_{x,y}(t))[y - x].$$

This derivative is continuous. Evaluation at $y - x$ is a continuous linear functional on E^*, hence commutes with the Bochner integral. The fundamental theorem of calculus therefore yields

$$\begin{aligned} \overline{D}H(x, y)[y - x] &= \int_0^1 DH(\gamma_{x,y}(t))[y - x]\, dt \\ &= H(\gamma_{x,y}(1)) - H(\gamma_{x,y}(0)) = H(y) - H(x). \end{aligned}$$

When $y = x$, the path is constant and the integral over an interval of length one is $DH(x)$, proving consistency. □

Theorem 2 (Exact AVF energy preservation)**.** *Let $J : E^* \to E$ satisfy*

$$g[J(g)] = 0 \qquad \text{for every } g \in E^*. \tag{3}$$

If $h \in \mathbb{R}$ and $x, y \in E$ satisfy the AVF step

$$y - x = hJ(\overline{D}H(x, y)), \tag{4}$$

then $H(y) = H(x)$.

Proof. Put $g = \overline{D}H(x, y)$. Equations (1), (4), and (3) give

$$H(y) - H(x) = g[y - x] = g[hJ(g)] = h\, g[J(g)] = 0.$$

□

For the usual Hamiltonian system $\dot{z} = S\nabla H(z)$ in $\mathbb{R}^d$, take a constant skew-symmetric matrix S and identify covectors with vectors. Then $J(g) = Sg$ satisfies (3), because $g^T Sg = -g^T Sg$. Linearity of S makes (4) precisely

$$y - x = h \int_0^1 S\nabla H(x + t(y - x))\, dt,$$

the average-vector-field method. Its energy conservation follows for every energy under the stated differentiability hypotheses and every step that solves the implicit equation. The theorem asserts preservation, not existence or uniqueness of that implicit solution. Replacing the integral by an inexact quadrature or solving the step inexactly requires a separate error analysis and is not the exact method in the source.

Formal verification and scope

The Lean development uses Mathlib's actual continuous linear maps, `HasFDerivAt`, `fderiv`, and Bochner interval integral. The main proposition universally quantifies over energies on any real normed space, assumes differentiability and continuity of the actual Fréchet derivative, and states the conjunction of the discrete-gradient identity, diagonal consistency, and preservation of every AVF step with (3). The main theorem is

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The proof establishes the chain rule along the actual affine segment, commutes evaluation with the actual integral, and applies Mathlib's fundamental theorem of calculus. No finite examples, numerical integration, or surrogate energy-difference formula replace these objects.

The finite-dimensional Riesz identification and the short matrix argument that skew-symmetry implies (3) are presented on paper. Lean proves the stronger coordinate-free preservation theorem directly under (3). These are the only bridges from the formal theorem to the usual gradient-and-skew-matrix notation. Both the identity and its averaging realization are fully formalized, as is the entire conservation argument. The allowed axioms printed by Lean are only `propext`, `Classical.choice`, and `Quot.sound`.

Source and reproduction

The unmodified bilingual statement is supplied as `SOURCE.md`; its frozen provenance and SHA-256 are recorded in the verification folder. The coordinating workflow found no prior submission for this conjecture. Lean is pinned to 4.19.0 and Mathlib commit

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

In the supplied Lean folder run

```
lake env lean -DwarningAsError=true Main.lean.
```

This draft was developed by a delegated AI agent with a self-review; the coordinating agent reviews it separately. No external independent review is claimed.